EDMS 20-305-2

SPECIFICATION
FOR
THREE-PHASE
INDIRECT SMART METER

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CONTENTS

1. SCOPE

Many important business processes conducted by the electricity distribution companies (EDCs) require re-engineering to increase effectiveness and efficiency. An important enabler for this re-engineering is an advanced metering infrastructure (AMI). The functionality of the AMI can contribute to the business needs identified by EEHC. Smart meters is the key role in AMI. Smart meters shall communicate using field area network (FAN) telecommunications technologies such as PLC, PLC/RF, mobile wireless service 2G/3G / 4G technology, and RS 485. The choice of FAN telecommunication technology shall optimize system design, architecture, interoperability and total cost of ownership while meeting this specification.

This specification describes the minimum technical requirements for designing, manufacturing, testing, inspection, supply and delivery of three-phase indirect smart meters. The required meters shall be suitable for outdoor installations of the power networks of _ _EDC.

2. APPLICABLE STANDARDS

- The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1).
- Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.
- Nothing in this specification shall lessen the bidder obligations in any other documents forming part of the contract.

Table (1)

Standard No.	Description
IEC 62052-11	Electricity metering equipment (a.c.) – General requirements, tests and test conditions – Part 11: Metering equipment
BS EN50470-1	Electricity metering equipment (a.c.) - General requirements, tests and test conditions. Metering equipment (class indexes A, B and C)
IEC 62055-21	Electricity metering – Payment systems – Part21: Framework for standardization
IEC 62055-31	Electricity metering – Payment systems – Part 31: Particular requirements- Static payment meters for active energy (classes 1 and 2)
BS EN50470-3	Electricity metering equipment (a.c.) - Particular requirements. Static meters for active energy (class indexes A, B and C)
IEC 62053-21	Electricity metering equipment - Particular requirements – Part21: Static meters for AC active energy (classes 0,5, 1 and 2)
IEC62053-22	Electricity metering equipment - Particular requirements – Part22: Static meters for AC active energy (classes 0,1S, 0,2S and 0,5S)
IEC 62053-23	Electricity metering equipment - Particular requirements - Part 23: Static meters for reactive energy (classes 2 and 3)
IEC 60060	High-voltage test techniques
IEC 60068	Environmental testing
IEC 62053-52	Electricity metering equipment (AC) - Particular requirements - Part 52: Symbols
IEC 60529	Degrees of protection provided by enclosures (IP Code)
IEC 62054	Electricity metering (a.c.) – Tariff and load control
IEC 62056-21	Electricity metering: data exchange for meter reading, tariff and load control – Part 21: Direct local data exchange
IEC 62056-51	Electricity metering data exchange for meter reading tariff and load control – Part 51: Application layer protocols
IEC62056-6-1	Electricity metering data exchange - The DLMS/COSEM suite - Part 6-1: Object Identification System (OBIS)

3. ENVIRONMENTAL CONDITIONS

– The electrical and mechanical properties of the required meters should be guaranteed under the environmental conditions given in Table (2).

– The meters shall comply the environmental requirements as defined in IEC 62052-11, IEC 62053-21 and IEC 62053-23. Additionally, the following requirements shall be met:

- The meter shall be protected against malfunction due to the ingress of vermin, dust, or excess humidity by conformal coating of the printed circuit boards in the meter
- Any openings shall be as small as practically possible to prevent the ingress of dirt and vermin and to limit the potential for vandalism or tamper following meter case

Table (2)

Ambient temperature	Maximum ambient temperature: 70 °C Minimum ambient temperature: -5 °C Maximum daily average temperature: 35 °C Maximum annual average temperature: 30 °C
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. DESIGN CRITERIA

4.1 Technical Requirements

– The meters shall comply with the technical requirements given in Table (3).

Table (3)

No.	Description	Specification
1	Rated voltage	Low Voltage LV Operated Network: Three Phase Meters: 3x220/380V
2	Variation in voltage	+30% to -40%
3	Continuous over voltage	Low voltage meters shall withstand 400 volts (Phase to neutral) for 48 hours without being damaged or causing changes of more than 0.01 kWh in its credit registers (excluding the possible decrement of credit due to power being consumed).
4	Rated frequency	50 Hz
5	Variation in frequency	± 5%
6	Power factor	Lag – unity – lead
7	Power consumption in the voltage circuit	10VA (2W) for voltage circuit. 1VA for phase current circuit for transformer operated meters Note: the above value for power consumption is for meter without communication modem.
8	RTC	Built-in real time clock with 10 years of operation battery backup
9	Display	LCD: Visibility shall be sufficient to read the meter mounted at height of 0.5 to 2.0 m. Pin type: trans-reflective HTN or STN type industrial grade; temperature range – 20 °C to +70 °C Meter LCD Messages shall be available in Arabic language.
10	Digits	WxH: 4 mm x 8 mm (minimum) with backlight.
11	Maximum viewing angle	160 degrees
12	Memory	Nonvolatile memory (EEPROM) that retains information up to 20 years without being damaged or corrupted during the meter operation.
13	Lightning protection	In accordance to IEC 62052-11, IEC 62053-21
14	Pulse output	Flashing LED visible from the front along with auxiliary 2-wire terminal for : - Active and Reactive energy consumption rate

– Indirect meters shall meet the electrical requirements given in Table (4).

Table (4)

No.	Description	Specification
1	Meter Type	Three-phase, four-wire, transformer-operated meter, active and reactive energy, four quadrant measurements
2	Number of elements	Four current measuring elements
3	Rated current, In	CT operated: 1A
4	Maximum continuous current, Imax	CT operated: 6A
5	Starting current	0.1% In according to IEC 62053-22
6	Class index	Class 0.5 for active energy according to IEC 62053-22 Class 2 for reactive energy
7	Number of display digits	10 including floating decimal point Number of cycles should be displayed
8	Display parameters	i. kWh for the current billing period (and previous months) ii. kVARh for the current billing period (and previous months) iii. Cumulative total kWh used since installation iv. Cumulative total kVARh used since installation. v. Last month consumption. vi. Date and time vii. Instantaneous voltage for each phase viii. Instantaneous primary current for each phase ix. Instantaneous power factor for each phase x. Average monthly power factor for each phase xi. Maximum demand (kW, Amp.) and its date andtime xii. Pervious maximum demand and its date andtime xiii. C.T. Ratio xiv. Panel breaker mode (auto or manual)
9	Power/ Load limiting	Meter shall disconnect the load when a pre-programmed threshold current is reached. The threshold shall be programmable/ configurable.
10	Limit of Current	Shall be programmable
11	Load disconnection	External contactor(breaker) controlled by auxiliary output signal from the meter

4.2 Meter Case

- The meter shall be effectively sealed to prevent entrance of moisture, rain and dust into its internal part with the degree of protection IP54 for outdoor installations.
- The meter body shall be a nonflammable and high impact, strong material.
- The meter base, body, and frame including terminal block shall be of heat resistive; shock proof and rust proof good quality hard material. (Compatible with IEC 60695-2).
- The meter base will be provided with three screw mounting holes, one slotted meter support bracket at the top and one round hole on each side in the bottom half of the base for securely mounting the meter to the meter board.
- The meter cover shall be made of polycarbonate material. The meter cover shall be provided with a window of polycarbonate or toughened glass.

4.3 Meter Sealing

- Provision shall be for sealing the meter with at least two Ferrule stainless steel seals
- The stainless steel seals shall be applied in such a way that it will not be possible to undo/loosen the mounting screws used to secure the meter without breaking these seals.
- The stainless steel seals shall be applied in such a way that they will be easily visible when viewing an installed meter from the front.
- The terminals shall also have Ferrule/TT sealing arrangement.

4.4 Meter Display

- Meter shall display all measured quantities or desired quantity automatically or scrolling via button on the meter as follows:
    - Active energy in each quadrant
    - Reactive energy in each quadrant
    - Active energy in first and fourth quadrant

- Reactive energy in first and second quadrant
- Monthly and quarterly maximum demand with exact time and date
- Time and date
- Serial number
- Battery failure
- The number of voltage cut offs and number of reverse energy events
- C.T. ratio
- Phase angle between current and voltage for each phase.
- instantaneous and average power factor

– Meter shall display and record the maximum demand with the ability to configure and calculate its time interval calculation. Maximum demand shall be displayed in three integer and two decimal digits at least.

– LCD shall display measured items in OBIS codes.

– Meter shall display and record temper events

– Meter display shall support Arabic language Interface.

4.5 Fault and Status Display

– The meter shall provide a visible indication of the incoming supply status.

4.6 Status/Alarm/Event Displays

– In addition to the displays required above, the meter must display the following minimum information in an intuitive way such that an inexperienced user can understand and interpret the information intelligibly (laminated instruction cards are to be provided for each unit):

- Meter ID
- Out of Credit message (for prepayment meters only)
- Meter tamper state
- Meter failure and/or fault code

– The bidder must submit a detailed specification of the way in which the displays are structured and laid out.

4.7 Meter Terminal

- The terminal block shall be made from fire-resistant material (polycarbonate grade 500R or equivalent Bakelite) materials.
- The terminal cover shall be made from fire-resistant material (polycarbonate), and shall be extended type. The meter terminals shall have two flat-head screws for wiring connection, these screws shall be made of brass. Furthermore, the meters shall meet the requirements given in Table (5).

Table (5)

No.	Description	Specification
1	Terminals materials	Brass or copper current terminals
2	Terminal bore diameter	3.5 mm

5. FUNCTIONAL REQUIREMENTS

5.1 Local Communication Interface

- The meter front shall have an IEC 62056-21 compliant optical communication port. This shall allow _ _ EDC to access the meter for a variety of interactions via a hand held unit (HHU).
- The meter shall have an additional 2-wire serial communication port

5.2 Remote Communication Interface

- The meter shall have remote communication option which will be able to interface with RS-485 and RJ-45 port for data communication with the central server from meters via GSM network with at 2G/3G/ 4G communication.
- The communication modules works as a plug-and-play model, without the need to modify/change the meter firmware.

– The communication protocol shall be compatible with the IEC 62056 DLMS/COSEM Application layer Protocol. Alternative open international standard solutions for the communication protocol will be considered on merit.
– The DLMS/COSEM profile(s) will be defined, programmed, and tested during the proof of concept and pilot deployments. These definitions, programming, and testing shall be at no additional cost to _ _ EDC.

5.3 Firmware Upgrade

– _ _ EDC shall be able to update / change the meter firmware locally via the optical port or other ports installed on the meter
– EDC shall be able to update / change the meter firmware remotely.
– The firmware upgrade mechanism shall be designed so as not to alter in any way the metering characteristics (metrology) or any of the following stored data:
  - Meter data
  - Meter status
  - Configuration parameters or operational parameters

  All of these data must remain unchanged even after firmware upgrades.
– New firmware will be submitted to the meter with date and time parameter of executing new firmware operation (the meter will upload the new updated firmware but will only execute it when the defined date and time parameter achieved).
– The meter will record the date and time of receipt of new firmware in the event log, as well as the date and time of the firmware upgrade process execution.
– The meter will perform self-check process after the execution of the new firmware update, and the result of self-check process will be stored on the meter event log,

- and shall be retrievable locally and remotely.
- The meter shall have non-volatile memory equal to or greater than ten times the size of the operational/running firmware image.
- The meter shall support two concurrent firmware images: current operational/running and pending upgrade

5.4Security

- The meters shall use a method for encryption/decryption of data exchanged in all meter interfaces, according to cyber-security department penetration test.
- A method employing minimally AES128-CBC/GCM shall be used in all information exchanges in private network, between meter and HES (Field Area Network, or FAN interface), and between the meter and hand held unit (HHU) via the optical interface. Enough reserve capacity in the components of the system (e.g., main processor, memory) shall exist to support AES256-CBC/GCM during the service life of the equipment. The meter shall support encryption and decryption of three-meter firmware images: current operational/running, pending upgrade, last successfully installed operational/running.
- The meter should be capable of supporting AES256-CBC/GCM during its operational lifetime without a hardware change.
- A method employing at least ECC256 shall be used in all information exchanges on public network as well as between meter and central system directly

- The bidder shall describe the security mechanism, including encryption/decryption methods, for the exchange of secure tokens or credits while the meter is in prepayment mode.
- The security methods and hardware shall be described in details by bidder. The

bidder shall prove following items:

- Security of method
- Client ability to change security key remotely
- Confidentiality of remote key changing
- If client changes the security key there is no way for meter manufacturers to retrieve new key (new key is not accessible for meter hardware, firmware, and external software)
- All input and output information decrypted and encrypted

5.5 Operational Modes

– The meter shall support at least two modes of operation: Normal mode and test mode.

5.6 Test Mode

– The test mode of the meter shall be activated using the optical port.

– In test mode, there shall be an automated test sequence available that includes:

- Full diagnostic test, that testing of the all the active and inactive functionality
- Metering accuracy test (enabled to allow for accuracy testing to be performed without affecting the recorded customer registration for energy)
- Connection validation tests (three-phase sequence connection).

– The test mode shall have the following displayed information (scrolled sequence):

- Meter firmware version
- Meter firmware update date and time stamp
- Meter tampering events
- Meter status
- Current limit

5.7 Load Profile

– At least four programmable channels for:

a. Active interval energy (import and export)
b. Reactive interval energy (import and export)
c. Active cumulative energy (import and export)
d. Reactive cumulative energy (import and export)

- Adjustable time interval from 1 to 60 minutes.

- Meter shall store these parameters for at least 30 days.

– At least four programmable channels for voltage and current as well as active and reactive power. Measurement type for each channel can be chosen as follows:
  a. Average
  b. Minimum
  c. Maximum

- Adjustable time interval from 1 to 60 minutes.
- Meter shall store these parameters for at least 30 days.

– Time and date shall be included in load profile.

– Load profile shall be readable in complete form and in defined blocks (based on start and end time and channels) locally and remotely.

5.8 Historic Consumption

– For this option, twelve energy registers shall store the total energy consumed for the previous twelve months. These registers are cycled at during the month-end process, with the oldest data being deleted.

5.9 Date and Time Management

– The meter shall be equipped with a real time clock supporting the date and time. Pseudo-clock using mains crossing detection is not acceptable. The maximum drift of the clock shall be less than 30 seconds per month. The battery shall operate maintenance-free for a period of at least 10 years.

– The meter shall be equipped with a battery monitoring system that monitors the battery's condition and sends a "Change Battery Alarm" message to the HES when the battery approaches the end of its operating life.

- When first commissioned into the AMI system the meter shall synchronize its time and date with that of the HES. Meter time and date synchronization shall also be done on a periodic basis through the HES, with a programmable periodicity, and done automatically during meter installation stage.

5.10 Power Quality

- Meter shall record under voltage, overvoltage, and power cut.
- Events related to under voltage and overvoltage shall be recorded in the meter. Threshold of under voltage is from 220 V down to 132 V by five volts decrements.For overvoltage,the threshold is from 220 Vup to 286 V by five volts increments
- The voltage events will not be recorded unless they continue for a time equal to or greater than the programmable time set for under voltage and overvoltage threshold. This time shall be adjustable / programmable between 1 to 60 seconds, in one-second increments.
- For each under voltage, minimum voltage occurred during the adjustable / programmable (1 to60) seconds period, the meter shall be detected and recorded as such; date and time of last occurrence and the cumulative number of the occurrences.
- For each overvoltage, maximumvoltage occurred and during the adjustable / programmable (1 to60) seconds period the meter shall be detected and recorded as such; date and time of last occurrence and the cumulative number of the occurrences.
- Parameters related to threshold and duration of under voltage and overvoltage shall be adjustable locally and remotely.
- Events related to power cut shall be recorded in the meter as follows:
  - For long duration power cuts (more than 3 minutes), duration time and date shall be recorded.
  - For short duration power cuts, number of power cuts shall be recorded.

– The AMI gather the following additional smart meter data

- Voltage at each phase (instantaneous and historical)
- Current at each phase (instantaneous and historical)
- Active power (instantaneous and historical)
- Reactive power (instantaneous and historical) Alarm and Event data
- Power outage data (occurrence/restoration time, etc.)
- Several event data such as overvoltage, voltage drop, current imbalance, etc.

5.11 Event Detection

– The meter shall classify and record the following commonly occurring events with date and time stamp. The tampering event log shall record any combination of the possible tamps given in Table (6).

Table (6)

No.	Event case	Indirect meter
1	Current Reverse	Phase current reverse (L1, L2, L3) in separate or combination.
2	Current imbalance	$I1 + I2 + I3 \neq In$ (Vectorially)
		Neutral current =0
3	Voltage missing	For any phase (L1, L2, L3)
4	Current missing	For any phase (L1, L2, L3)
5	External magnetic field influence	✓
6	Breaker mode changing to manual	✓
7	Opening meter cover	✓
8	Opening panel door	✓
9	Opening terminal cover	✓
10	Communication failure events	✓

– There shall be a mechanism by which the consumer can be disconnected immediately when a verifiable tamper event occurs.

- The event status shall be reset by using the meter optical port and send the reset event to the HES time stamped.
- The event status shall be reset remotely.
- The meter shall sends a push message to the HES with every Event, stamped with date and time.
- The meter shall provide EVENT log data with the last occurred EVENT date and time stamp and it can be easily readable remotely or locally.

5.12 Meter Metrology

- The meter shall have the ability of measuring and recording import/export active energy
- The values to be recorded for import and export are actual values.
- The meter should have the ability of measuring and recording import/export reactive energy (the values to be recorded for import and export are actual values).
- The meter shall measure phase current (True RMS) in A
- The meter shall measure phase voltage (True RMS) in V
- The meter shall measure power factor
- The meter shall measure load profile
- The measured energy quantity shall have the following resolutions:
  - Resolution of active energy is 0.01 kWh or better.
  - Resolution of reactive energy is 0.01 kVARhor better.
- It shall be possible to locally and remotely select or configure exported energy whether to be registered or not.(According to standard, when the energy flows from network to the consumer, it is registered as IMPORT. When the energy flows from consumer to the network, it is registered as EXPORT.) Meter has the ability to add or not the consumed exported energy to the imported energy as total consumed

energy.

- The meter is configured in such way not to register the consumed energy as Export (exceptDistributed Generations). Whenever it is occurred, the system shall record this event (in case of DG the meter register only exported energy and in this case not added to imported energy).
- The meter shall measure import/export energy as well as Absolute value of active energy and it shall be possible to activate this capability.
- Measuring and recording the maximum demand of active power shall be based on configurable time intervals and subintervals (sliding mode).
  - It shall be possible to reset maximum demand remotely
  - It shall be possible to reset maximum demand locally via sealed push button on front of the meter.
  - It shall be possible to automatically reset the maximum demand at specific billing period.

  Above capability shall be activated (or not) and all parameter shall be adjustable remotely and locally
- The meter shall have the ability to measure the angle between voltage phases.
- The meter shall have the ability to measure the maximum demand.
- The meter shall have the ability to give phasor diagram for the measured voltage and current (monitor and save it to meter computer interface program and/or HES)
- Note: units on active energy (Wh), reactive energy (VARh), and demand (W) shall be configurable as kilo or Mega.

5.13Tariff

- All tariff calculation and credit calculation shall be processed inside the meter.
- The meter shall support stepped tariffs structures as well as time-of-use and max.

demand.

– The tariff structure consists of the following records:

- Each tariff must be uniquely identified using the tariff code
- Each tariff shall have an activation date, being the date on which the tariff becomes effective.
- Each tariff has up to ten steps / time slot (in kWh) for different levels of energy pricing.
- Each tariff has up to ten steps / time slot (in kVARh) for different levels of energy pricing.
- The rate describes the cost per kWh for energy consumption in that slot.
- The rate describes the cost per kVARh for energy consumption in that slot.

5.14Tariff switchover

– When the meter detects that a new tariff is applicable (using the tariff activation date), the meter shall execute the following steps:

- The meter shall send the consumption to the MDM for billing.
- The current tariff code shall be updated to reflect the new tariff code.
- The old (expired) tariff should be deleted.
- New tariffs will only be activated at 00:00 on the first day of a month.

5.15 Billing

– Billing operates on a "monthly" basis (a period of days definable by _ _ EDC). The energy consumption charge is deducted within the meter for at least every 1 kWh. All other charges including service charge, minimum charge, VAT, and other taxes are deducted at the POS (point of sale).

6. MARKING

- Every meter shall be indelibly marked (engraved on the terminal cover surface) with a connection diagram. The meter terminals shall be marked and indicated on this diagram.
- Every meter shall have clearly visible, indelibly and distinctly marked name plate containing the following information:

  a. Manufacturer's name
  b. Meter type
  c. Number of phases and number of wire
  d. Meter serial number
  e. Year of manufacture
  f. Rated voltage of the system
  g. Basic current and maximum current
  h. Reference frequency in hertz
  i. Meter constant in imp/kWh
  j. Meter Constant in imp/kVARh
  k. Class index of the meter
  l. _ _EDC logo
  m. Contract No.
  n. Reference Standards

7. TESTING and INSPECTION

7.1 Routine Acceptance Tests

- The routine / acceptance tests, given in Table (7), shall be carried out on all offered meters, and shall be carried out by the manufacturer / contractor, with providing a factory acceptance test certificate for the manufactured meters.
- The tests shall be carried out in accordance with IEC 62053-21, IEC 62053-22, IEC 62052-11, BS EN 50470-1, BS EN 50470-3, and tender requirements.
- Number of samples selection criteria shall be done in accordance to the IEC 514 Standard.

Table (7)

No.	Test Name		As per Standard
1	Test of meter constant		IEC 62052-11, IEC62053-22 for active, IEC 62053-23 for reactive
2	Limits of error due to variation of the Current		
3	Test of no-load condition		
4	Test of starting condition		
5	Voltage variation		
6	Frequency variation		
7	Test of power consumption		
8	Influence of self-heating test(*)		IEC 62053-22 for active, IEC 62053-23 for reactive
9	Influence of ambient temperature variation(*)		
10	Influence of short time overcurrent(*) if applicable		
11	Resistance to heat and fire test(*)		
12	Visual inspection		
	1	Meter display	As per Specification
	2	Meter casing	
	3	Meter sealing	
	4	Meter name plate marking	
	5	Connection diagram and terminal marking	
	6	Packing	

13	Event test		
Three Phase Tests	1	Phase (I1 , I2 , I3)current and or neutral current reverse connection	As per Specification
	2	Current imbalance (I1+ I2 + I3 ≠ In) (vectorially)	
	3	Neutral current = 0	
	4	Missing phase voltage	
	5	External magnetic influence	
	6	Open meter cover and or terminal cover and panel door	
14	Functional tests		
	1	Meter mode	As per Specification
	2	Maximum demand	
	3	Load profile	
	4	Meter local communication (optical, RS485)	
	5	Meter remote communication	
	6	Load limit	
	7	Tariff	
	8	Tariff switchover	
	9	Friendly hours, weekend, holidays	
	10	Load control	

(*)The first random sample will be collected from the first delivered batch, and the other random sample will be collected from the rest of the delivered meters. In case of Sample test failure the delivered meter batch will be rejected, then collect other random sample from each later delivered batches in order to apply these tests, and the order will completely canceled in case of the failure of applying these tests three times.

Table (8)

No. of meters per batch (N)		First sample			Second sample	No. of meters for both samples (n1+n2)
		No. of meters per sample (n1)	**No. of rejected meters (d1)**	**No. of allowed rejected meters (c1)**	**No. of meters per sample (n2)**	**Total No. of allowed rejected meters (c2)**
1st Case	101-500	30	2	0	30	1
2nd Case	501-1000	40	2	0	40	2

7.2 Meter Type Tests

- The meter shall have a type test certificate that include details of test results according to IEC62052-11, IEC62053-21, IEC62053-22, IEC 62053-23, BS EN 50470-3from international accredited laboratory.
- The bidder shall provide a type test report issued within the last 5 years. (Submission is mandatory before the pilot)
- The type test must include all tests that applicable for that particular type in the standards. Minimally expected tests are given in Table (9).

Table (9)

NO	Test Name	As per Standard
1	Test of insulation properties	
1.1	Impulse voltage tests	IEC62052-11, EN50470-1
1.2	AC Voltage tests	
2	Tests of accuracy requirements	
2.1	Test of meter Constant	BS EN 50470-3 for active, BS EN 61268 for reactive
2.2	Test of Starting Condition	
2.3	Test of No-Load condition	
2.4	Test of influence quantities	
3	Test of Electrical Requirements	
3.1	Test of power consumption	BS EN 50470-3
3.2	Test of influence of short-time overcurrent	
3.3	Test of influence of self-heating	
3.4	Test of accuracy in the presence of harmonics	
3.5	Test of influence of supply voltage	IEC62052-11, EN50470-1
3.6	Test of influence of heating	
4	Test for Electromagnetic Compatibility (EMC)	
4.1	Test of immunity to electrostatic discharges	IEC62052-11, EN50470-1
4.2	Test of immunity to electromagnetic RF fields	
4.3	Fast transient burst test	
4.4	Test of immunity to conducted disturbances, induced by radio frequency Field	
4.5	Surge immunity test	
4.6	Damped oscillatory waves immunity test	
4.7	Radio interference suppression	

5	Test of effect of the climate environments	
5.1	Dry Heat test	IEC62052-11, EN50470-1
5.2	Cold test	
5.3	Damp heat cyclic test	
5.4	Solar radiation test	
6	Mechanical Test	
6.1	Vibration test	IEC62052-11,EN50470-1
6.2	Shock test	
6.3	Spring hummer test	
6.4	Tests of protection against penetration of dust and water	
6.5	Test of resistance to heat and fire	

7.3 DLMS Compliance Certificate

- The meter shall have a certificate for IEC62056 DLMS/COSEM, for the following: (Submission is mandatory upon submission of proposals)
  - Optical port Communication
  - 2-wire Serial RS485 Communication port
  - WAN / FAN Communication Modem
- Failure to submit any one of the theses certificates will be considered as a failing of the bidder to submit the whole item (DLMS Compliance Certificate).

7.4 Communication Modem Certificate

- Communication modem should have certificate for industrial equipment from aninternational accredited laboratory (Submission is mandatory before the proof of concept)

7.5 Lifetime Test Certificate

- The meter must be certified for 15 to 20 years life of operation, according to theAccelerated Lifetime (ALT) IEC testing procedures found in IEC62059-41, IEC62059-31-1, and IEC62059-32-1. (Submission is mandatory before the proof of concept).

8. SUBMITTALS

- Dimensional drawings of the offered meter including connection diagrams.
- Original catalogues shall be submitted to facilitate evaluation of the offer.
- Type test reports/certificates, DLMS compliance certificate, communication modem certificate, and meter life time test certificate from an independent testing laboratory shall be submitted to _ _ EDC.
- Approval certificate issued by Egyptian Electricity Holding Company.
- List of deviations and clauses to which exception is taken (if any).
- The contractor should provide complete installation, operation and maintenance instruction/manuals with each meter.
- All information should be clearly readable, with IEC standard symbols and SI units.
- The parameterization tool of meters should be delivered.